

SHADOW - Main Result

Calculation: 5 Old WTG and 2 New WTG

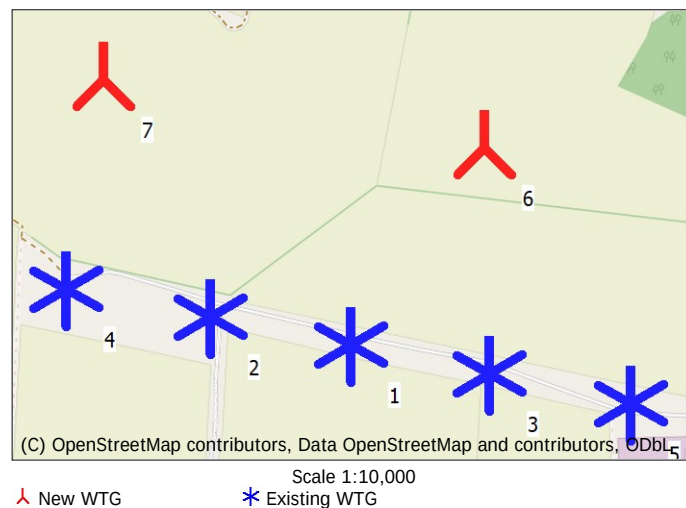
Assumptions for shadow calculations

Maximum distance for influence
Calculate only when more than 20 % of sun is covered by the blade
Please look in WTG table

The calculated times are "worst case" given by the following assumptions:
The sun is shining all the day, from sunrise to sunset
The rotor plane is always perpendicular to the line from the WTG to the sun
The WTG is always operating

A ZVI (Zones of Visual Influence) calculation is performed before flicker calculation so non visible WTG do not contribute to calculated flicker values. A WTG will be visible if it is visible from any part of the receiver window. The ZVI calculation is based on the following assumptions:
DHM: Elevation Grid Data Object: OS_Terrain_50_2020_30km_x_30km.wpg (1)

All coordinates are in
British TM-OSGB36/Airy (GB/IE)

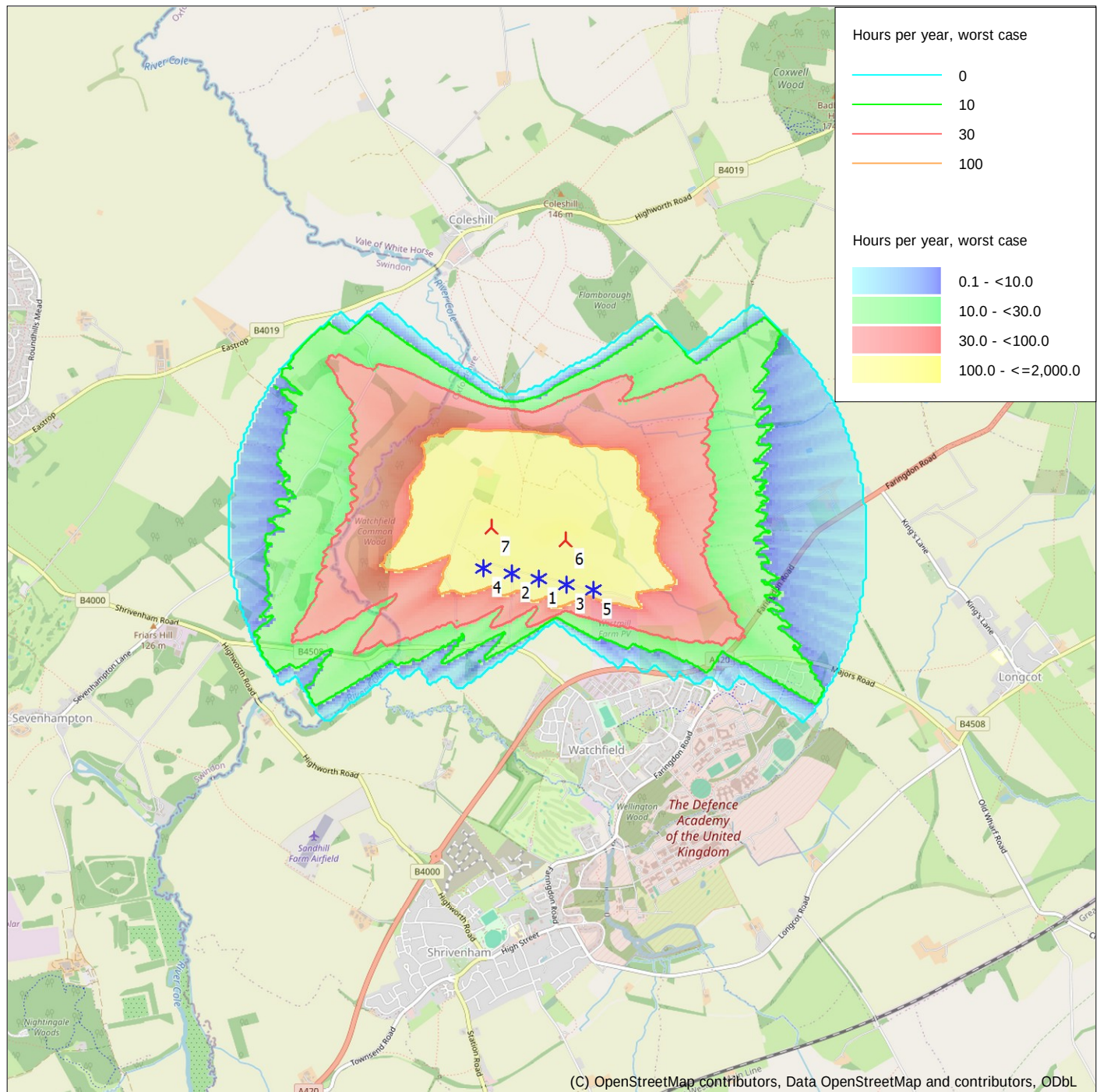


WTGs

	Easting	Northing	Z	Row data/Description	WTG type			Power, rated [kW]	Rotor diameter [m]	Hub height [m]	Shadow data	
					Valid	Manufact.	Type-generator				Calculation distance [m]	RPM
			[m]									
1	424,115	191,390	109.3	BONUS 1.3 MW 1300-250 62.0 !O! hub: 50.0 m (TO...	No	BONUS	1.3 MW-1,300/250	1,300	62.0	50.0	1,144	19.0
2	423,930	191,427	112.1	BONUS 1.3 MW 1300-250 62.0 !O! hub: 50.0 m (TO...	No	BONUS	1.3 MW-1,300/250	1,300	62.0	50.0	1,144	19.0
3	424,300	191,353	107.9	BONUS 1.3 MW 1300-250 62.0 !O! hub: 50.0 m (TO...	No	BONUS	1.3 MW-1,300/250	1,300	62.0	50.0	1,144	19.0
4	423,738	191,463	113.3	BONUS 1.3 MW 1300-250 62.0 !O! hub: 50.0 m (TO...	No	BONUS	1.3 MW-1,300/250	1,300	62.0	50.0	1,144	19.0
5	424,488	191,315	106.0	BONUS 1.3 MW 1300-250 62.0 !O! hub: 50.0 m (TO...	No	BONUS	1.3 MW-1,300/250	1,300	62.0	50.0	1,144	19.0
6	424,293	191,652	103.2	VESTAS V162-6.2 6200 162.0 !O! hub: 119.0 m (TO...	Yes	VESTAS	V162-6.2-6,200	6,200	162.0	119.0	2,044	-
7	423,787	191,740	113.4	ENERCON E-160 EP5 4600 160.0 !O! hub: 119.9 m (...	No	ENERCON	E-160 EP5-4,600	4,600	160.0	119.9	1,781	9.4

SHADOW - Map

Calculation: 5 Old WTG and 2 New WTG



Map: EMD OpenStreetMap , Print scale 1:40,000, Map center British TM-OSGB36/Airy (GB/IE) East: 424,120 North: 191,520
 New WTG Existing WTG
 Flicker map level: Elevation Grid Data Object: OS_Terrain_50_2020_30km_x_30km.wpg (1)
 Time step: 3 minutes, Day step: 7 days, Map resolution: 20 m, Visibility resolution: 10 m, Eye height: 1.5 m